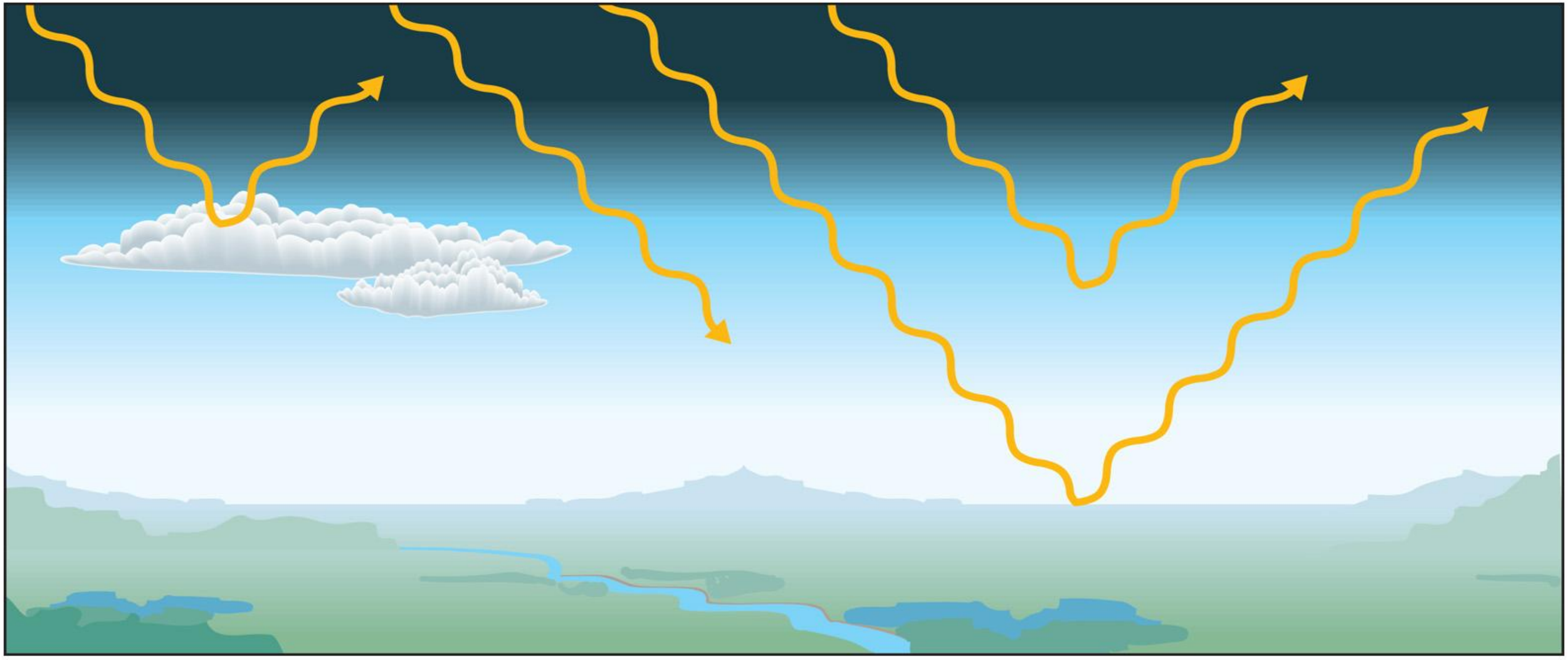
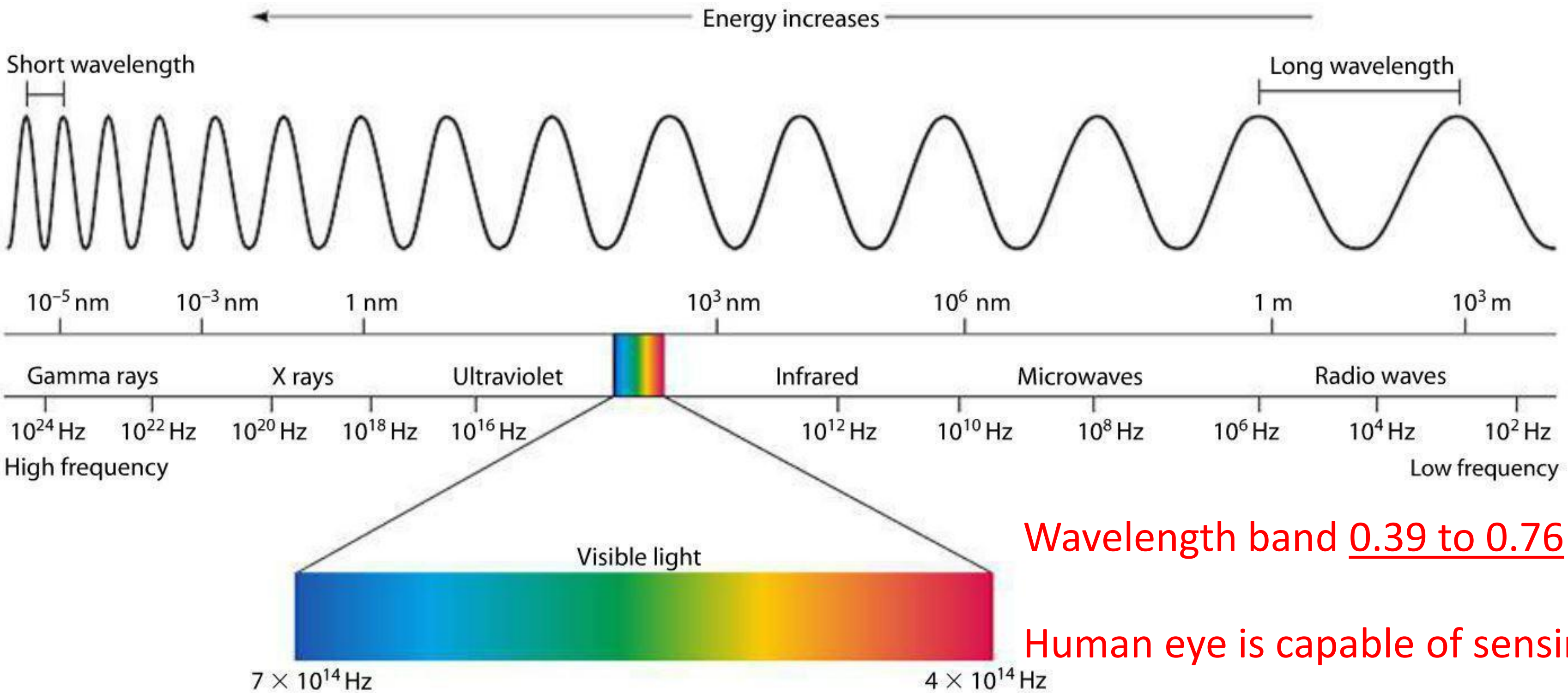


Atmospheric Radiation

Spectrum of the EM radiation:

- EM radiation can be viewed as an “ensemble of waves propagating at the speed of light”



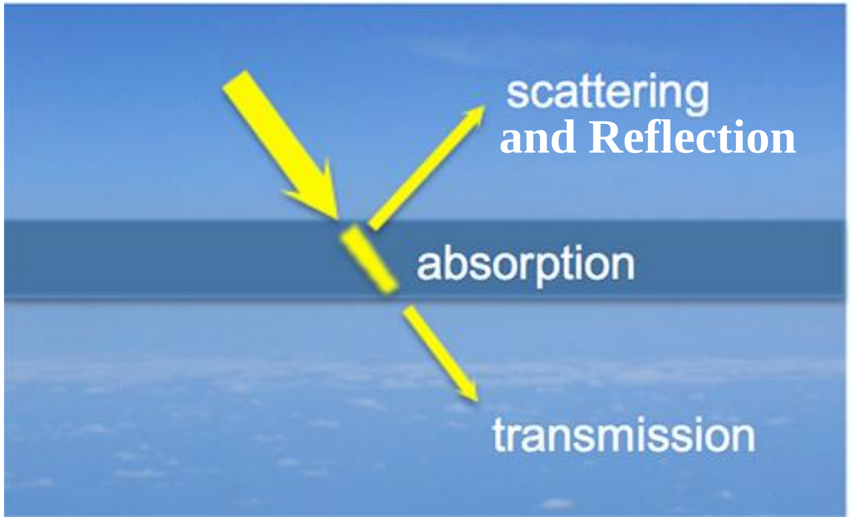
Wavelength band <u>0.39 to 0.76</u> μm

Human eye is capable of sensing.

<u>Shortwave radiation</u>, $\lambda < 4$ µm -- Energy associated with the Solar radiation

<u>Longwave radiation</u>, $\lambda > 4$ µm -- Energy associated with the Earth emitted radiation

Radiation and Matter



- When radiation encounters matter, four things can happen. It can be *transmitted through* matter; it can be *absorbed by* the matter; it can be *scattered by* the matter.

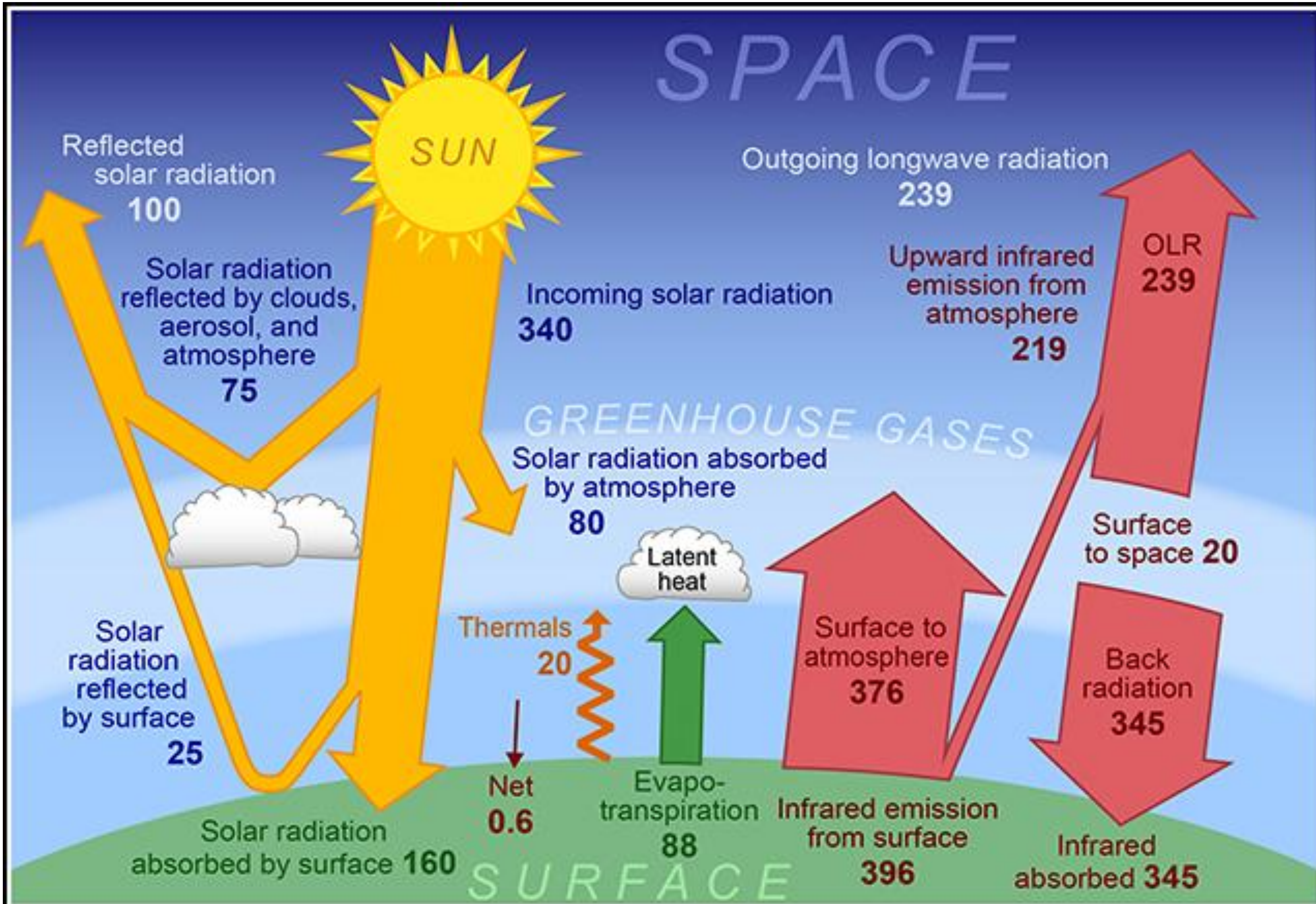
Absorption – the full energy transfer from radiation to a atmosphere, leads to heating. E.g. **UV radiation**– absorbed by O_3 (stratosphere)

Transmission – Radiation passes through a atmosphere without being absorbed or scattered. It does not affect the atmosphere.

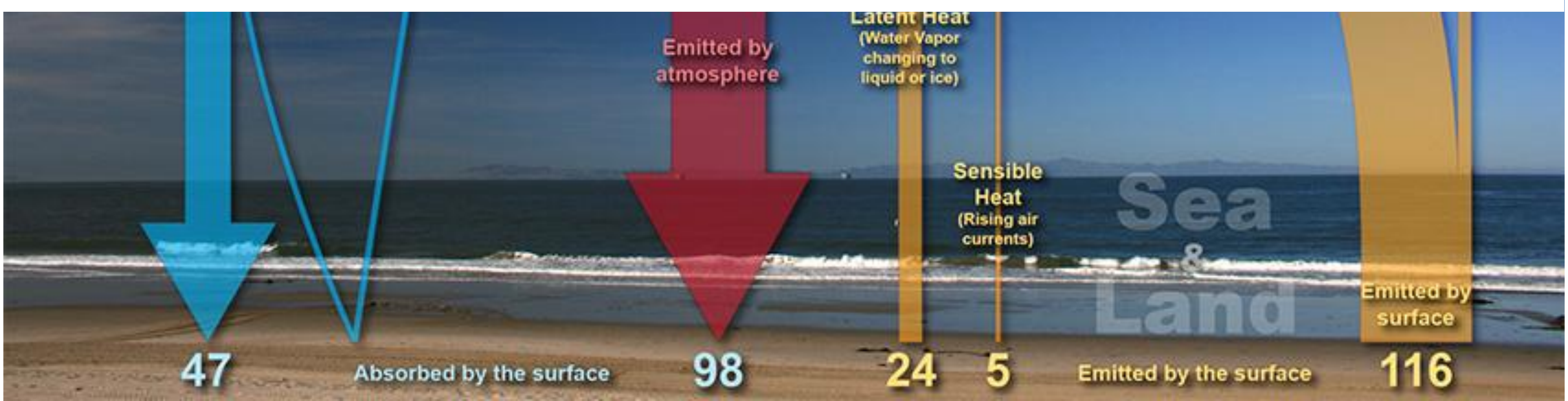
Reflection – describes the process whereby incident radiation bounces off the surface of a substance

Scattering – Radiation is reflected and scattered by the atmosphere, but is not absorbed

Global Energy Budget

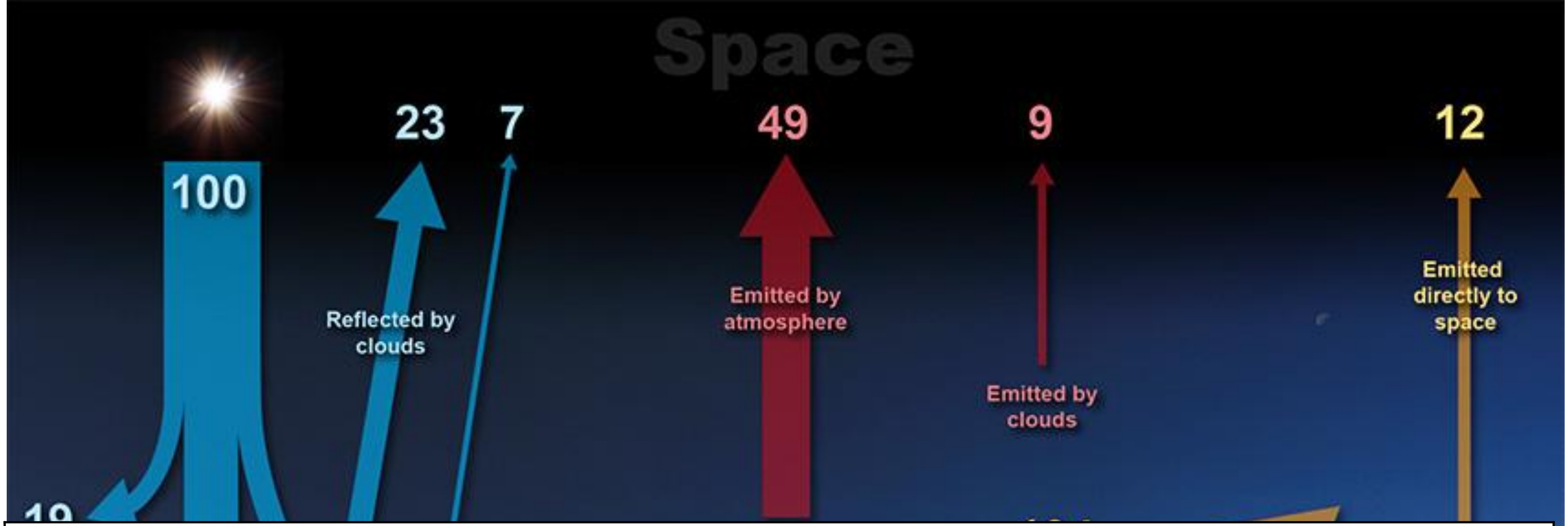


At the earth's surface - Energy absorbed is balanced with the energy released.			
Incoming energy		**Outgoing energy**	
Units	**Source**	**Units**	**Source**
+47	Absorbed shortwave radiation from the sun.	**-116**	Longwave radiation emitted by the surface.
+98	Absorbed longwave radiation from gases in atmosphere.	**-5**	Removal of heat by convection (rising warm air).
		-24	Heat required by the processes of evaporation and sublimation and therefore removed from the surface.
+145	Total Incoming	**-145**	Total Outgoing



The atmosphere itself - Energy into the atmosphere is balanced with outgoing energy from atmosphere.

Incoming energy		Outgoing energy	
Units	**Source**	**Units**	**Source**
+19	Absorbed shortwave radiation by gases in the atmosphere.	-9	Longwave radiation emitted to space by clouds.
+4	Absorbed shortwave radiation by clouds.	-49	Longwave radiation emitted to space by gases in atmosphere.
+104	Absorbed longwave radiation from earth's surface.	-98	Longwave radiation emitted to earth's surface by gases in atmosphere.
+5	From convective currents (rising air warms the atmosphere).		
+24	Condensation /Deposition of water vapor (heat is released into the atmosphere by process).		
+156	Total Incoming	-156	Total Outgoing



At the top of the atmosphere - Incoming energy from the sun balanced with outgoing energy from the earth.

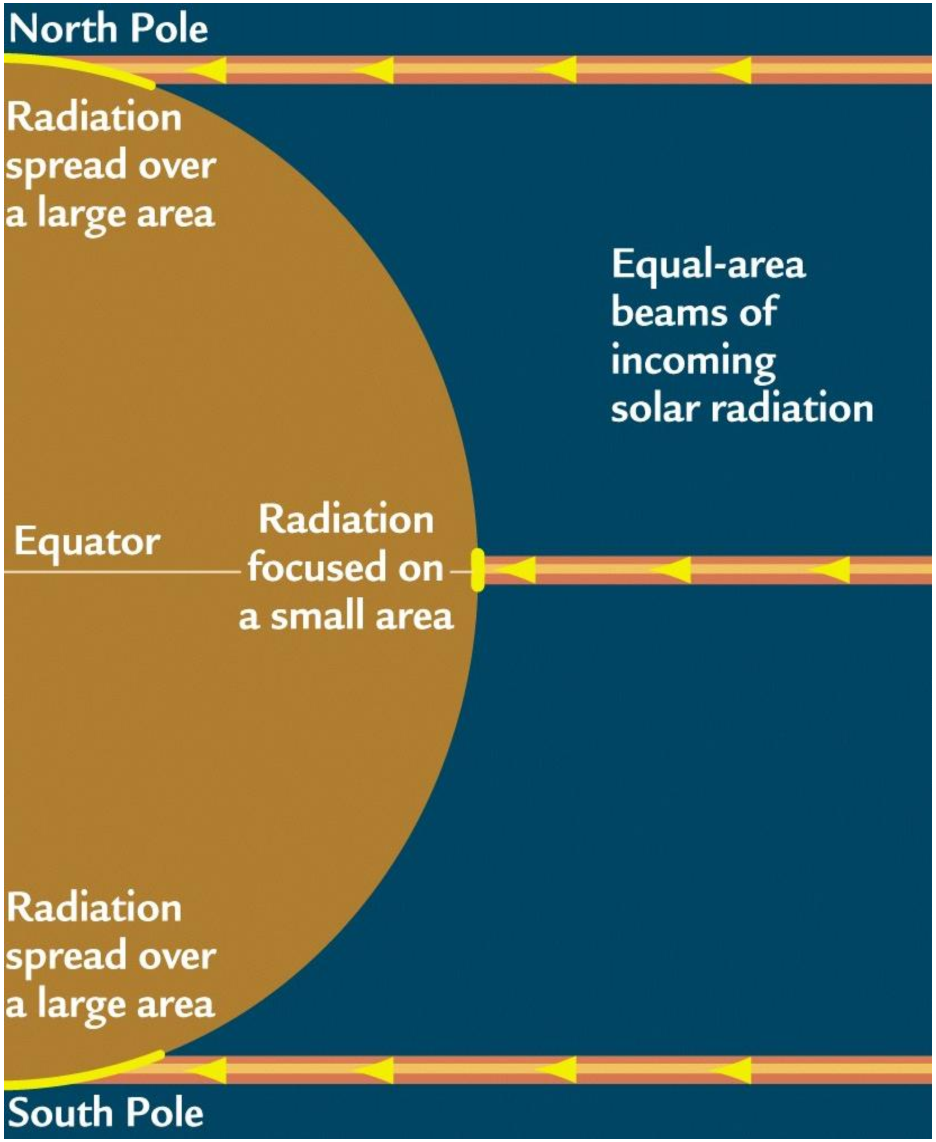
Incoming energy		Outgoing energy	
Units	**Source**	**Units**	**Source**
+100	**Shortwave radiation from the sun.**	**-23**	Shortwave radiation reflected back to space by clouds.
		-7	Shortwave radiation reflected to space by the earth's surface.
		-49	Longwave radiation from the atmosphere into space.
		-9	Longwave radiation from clouds into space.
		-12	Longwave radiation from the earth's surface into space.
+100	**Total Incoming**	**-100**	Total Outgoing

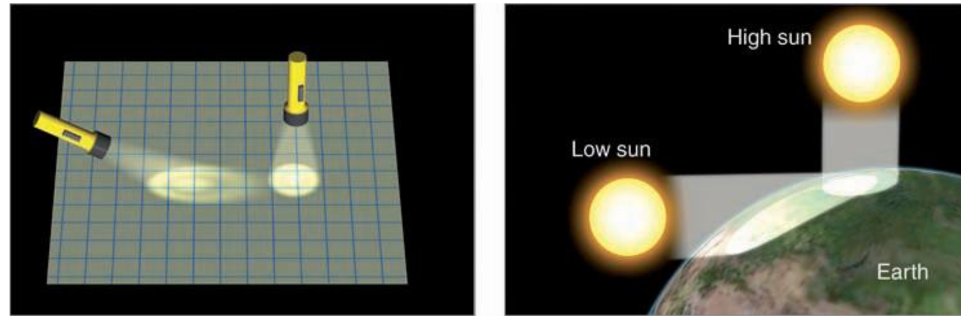
Variations in Heat Balance

☞ Incoming solar radiation

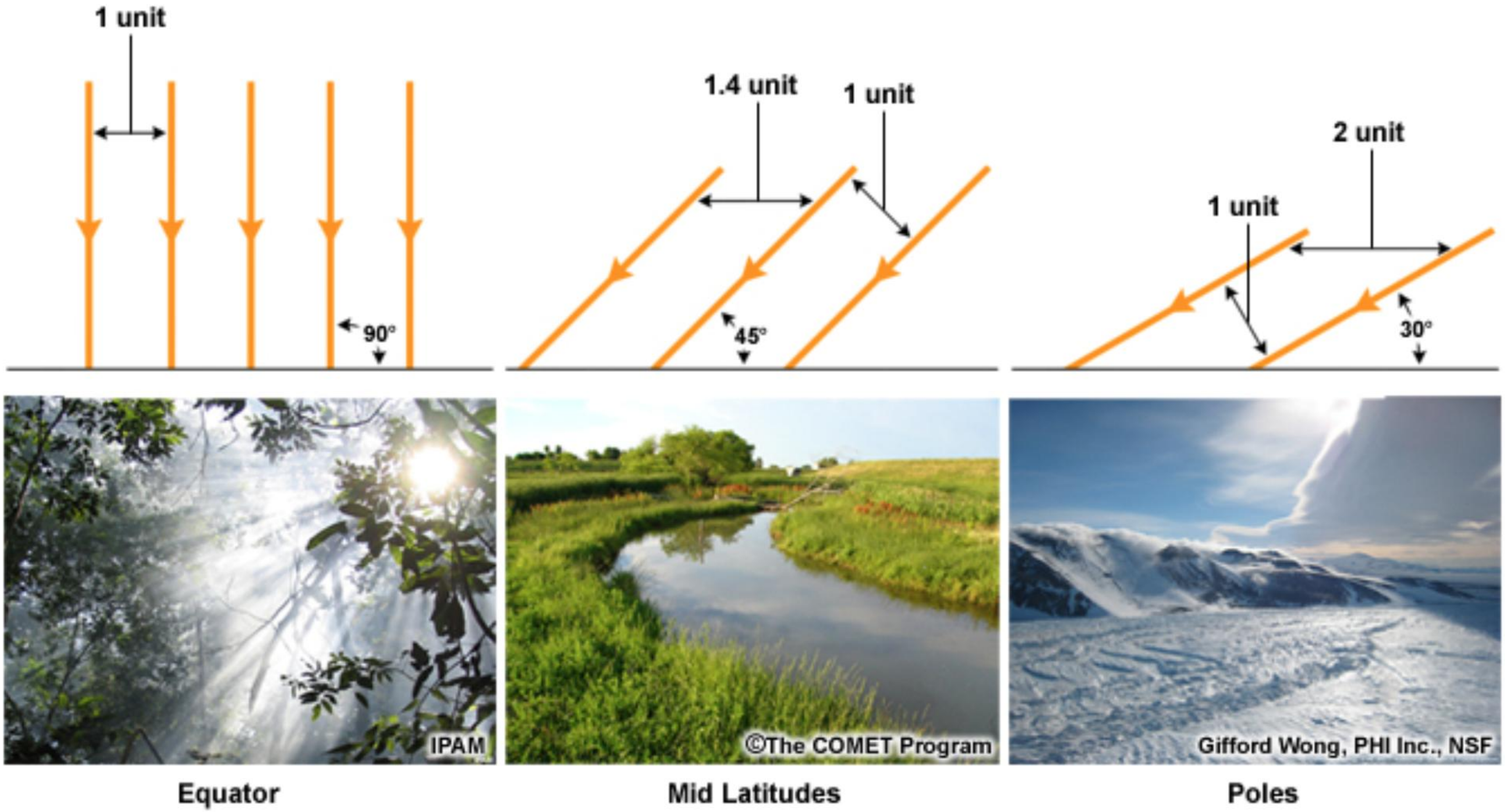
- Stronger at low latitudes
- Weaker at high latitudes

☞ Tropics receive more solar radiation per unit area than Poles.



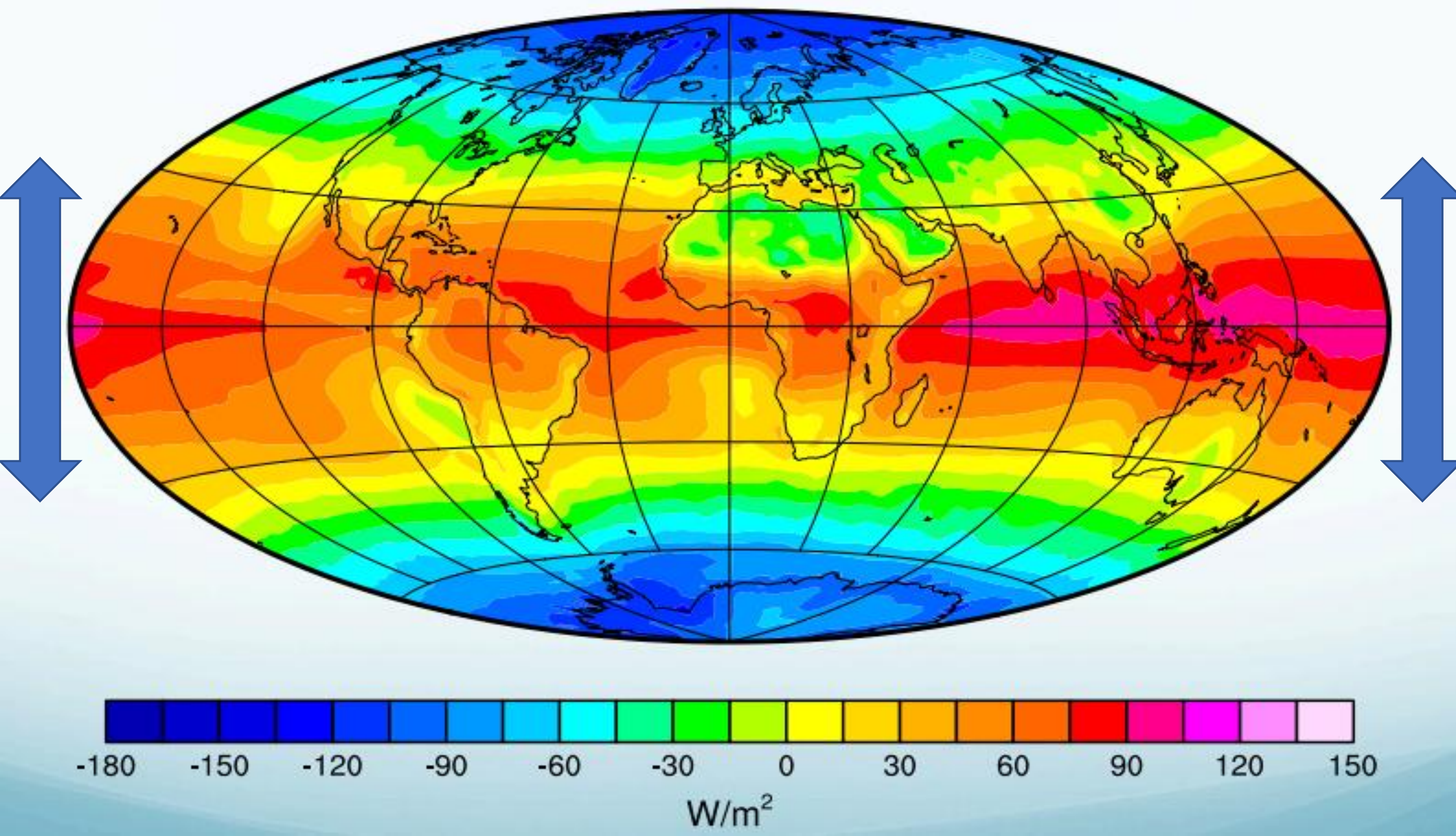

High sun
Low sun
Earth

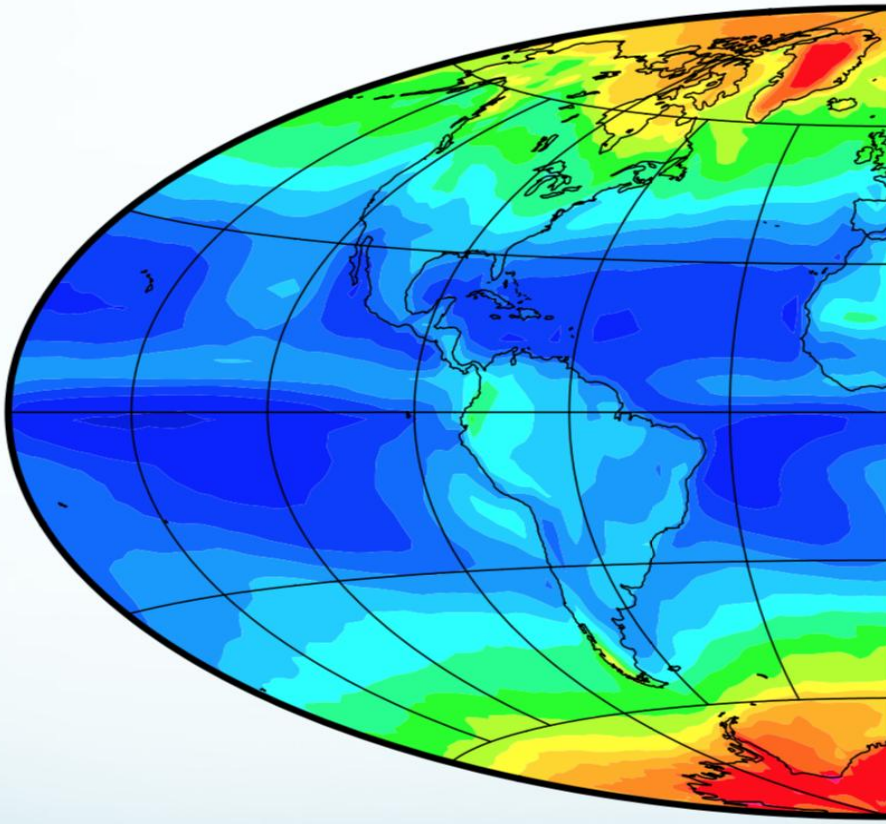
b


1 unit
90°
1.4 unit
1 unit
45°
2 unit
1 unit
30°
IPAM
©The COMET Program
Gifford Wong, PHI Inc., NSF
Equator
Mid Latitudes
Poles

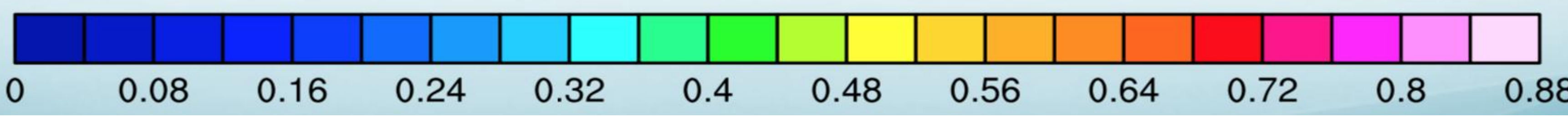
Net Radiation – Annual Mean

Absorbed - Outgoing



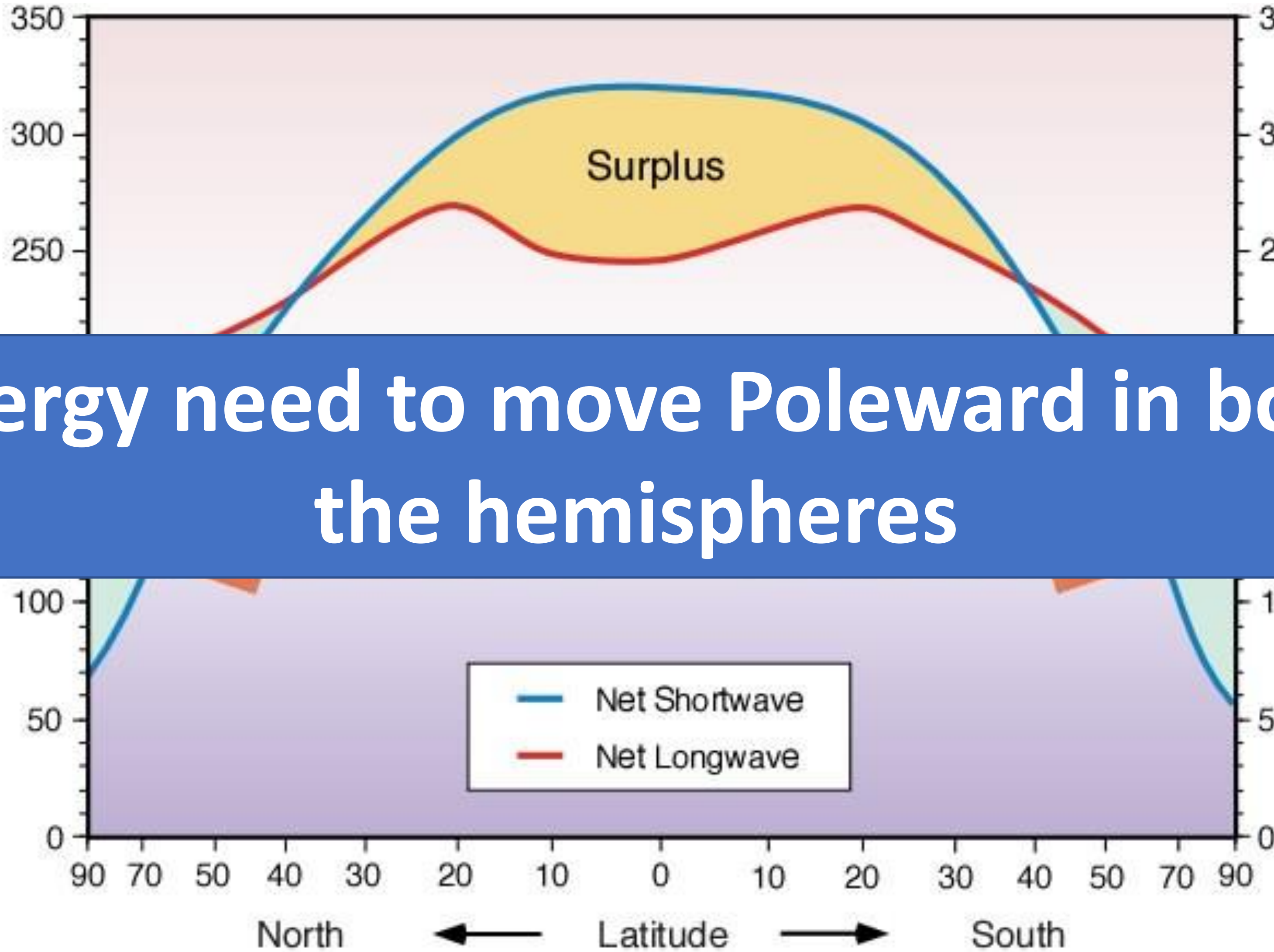


Type of surface	Albedo (%)
Ocean	2 – 10
Forest	6 – 18
Cities	14 – 18
Grass	7 – 25
Soil	10 – 20
Grassland	16 – 20
Desert (sand)	35 – 45
Ice	20 – 70
Cloud (thin, thick stratus)	30, 60 – 70
Snow (old)	40 – 60
Snow (fresh)	75 – 95



☞ **ALBEDO:** **The ratio of the amount of electromagnetic radiation reflected by the earth's surface to the amount incident upon it.**

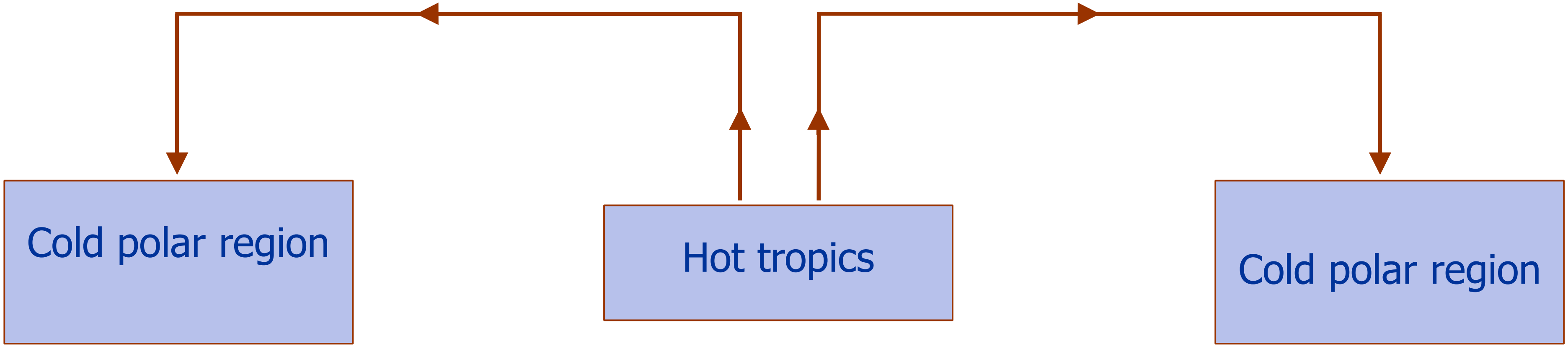
☞ **Value varies with the surface composition. For example, snow and ice vary from 80% to 85% and bare ground from 10% to 20%.**



Energy need to move Poleward in both the hemispheres

Net heat flux at the top of the atmosphere is Positive over Tropics and Negative over the polar region

How is equilibrium of temperature maintained By earth's atmosphere?



- Circulation is setup due to thermal gradients
- Transports heat from tropics to the polar region
- Large north-south temperature gradient also gives rise to some unstable planetary circulations. They also transport heat poleward

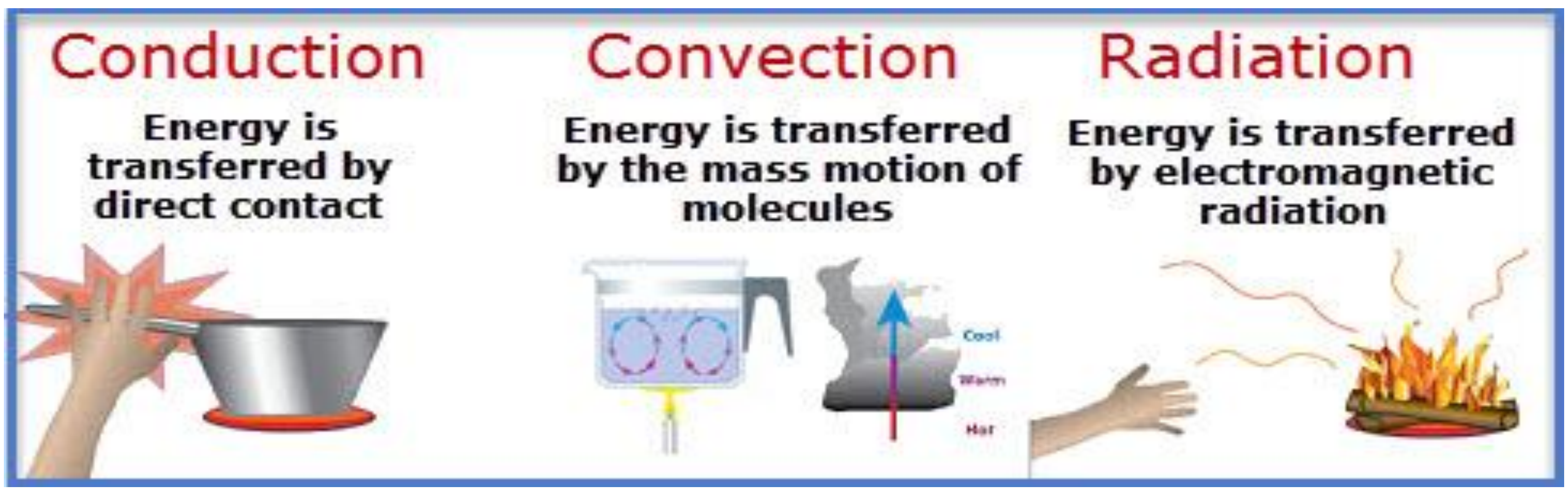
Heat Exchange Processes in the Atmosphere

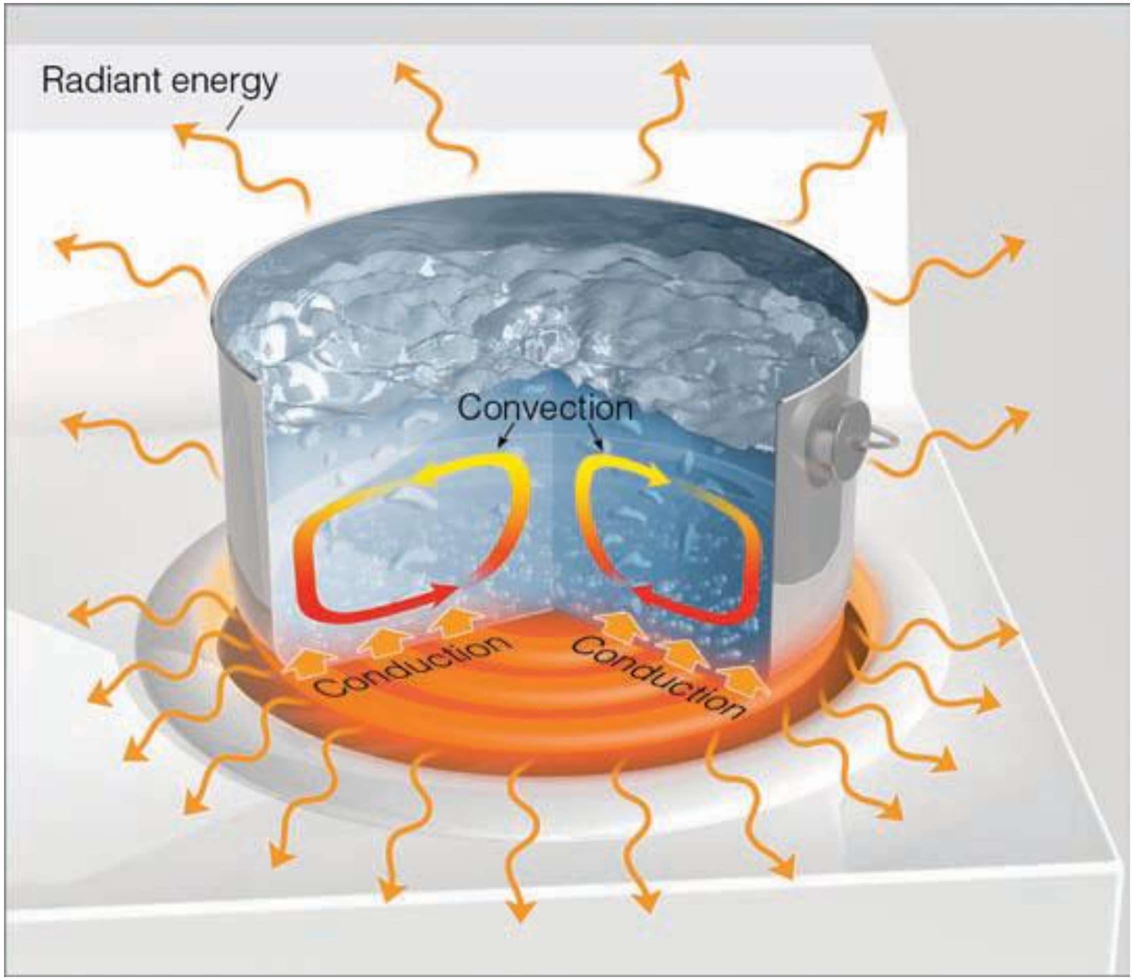
☞ Most important <u>form of energy </u>for weather and climate, receive from the sun - *<u>radiant energy</u>*.

☞ Energy, therefore, takes on many forms, and it can change from one form into another.

☞ Energy is transferred between the earth's surface and the atmosphere through:

Radiation Conduction Convection





The hot burner warms the bottom of the pot by conduction. The warm pot, in turn, warms the water in contact with it. The warm water rises, settings up convection currents. The pot, water, burner, and everything else constantly emit radiant energy (orange arrows) in all directions.

Radiation Laws (Blackbody)

☞ A blackbody is one which absorbs all incident radiation. A blackbody also emits the maximum possible irradiance at all wavelengths, for a given temperature.

☞ All objectives (whose temperature is above absolute zero), no matter how big or small, emit radiation.

I. STEFAN-BOLTZMANN LAW:

The radiation flux (amount of radiation emitted per unit surface area) of an object is proportional to the 4th power of its absolute temperature.

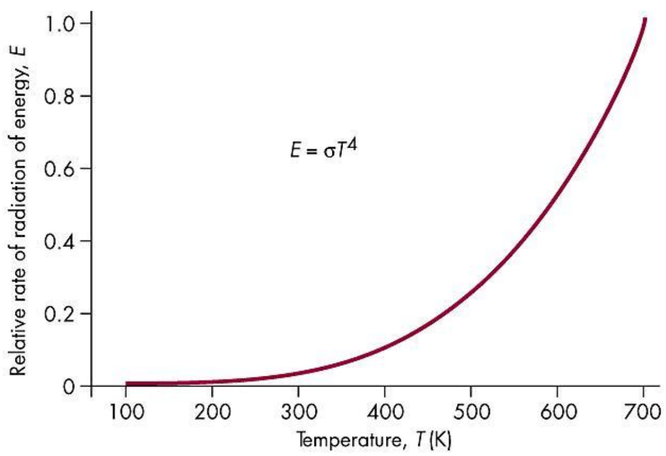
- When the temperature of an object increases, more total radiation is emitted:

$$\boxed{E = \sigma T^4}$$

E is the maximum rate of radiation emitted

T is the object's surface temperature

σ is Stefan-Boltzmann constant 5.67×10^{-8} W/m^2.K^4

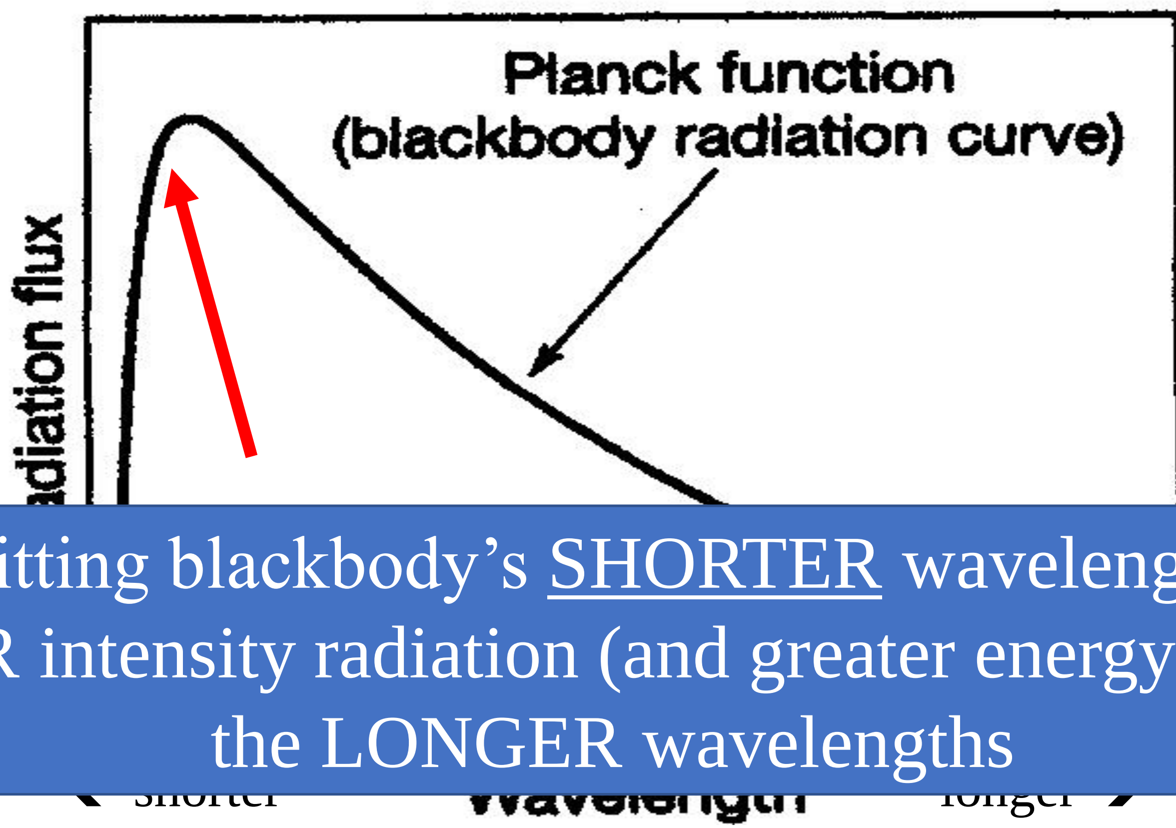


"The hotter the body, the (much) greater the amount of energy flux or radiation"

(II) PLANCK'S LAW : The Planck function describes the blackbody spectrum, that is the blackbody monochromatic radiance, $L_{\lambda B}$ (also denoted by B_λ in some books), as a function of temperature and wavelength:

$$L_{\lambda B} = \frac{2hc^2}{\lambda^5} \frac{1}{\exp(hc/\lambda kT) - 1}$$

where h is Planck's constant (6.625×10^{-34} Js), and k is Boltzmann's constant (1.38×10^{-23} JK^{-1}). The blackbody radiance can also be expressed as a function of frequency using the fact that



An emitting blackbody's <u>SHORTER</u> wavelengths have HIGHER intensity radiation (and greater energy flux) than the LONGER wavelengths

(III) Wien’s displacement Law:

Relationship b/w Temperature and Wavelength

As substances get HOTTER, the wavelength at which radiation is emitted will become SHORTER. This is called Wien's law.

Wien’s Law can be represented as:

$$\lambda_m = a/T$$

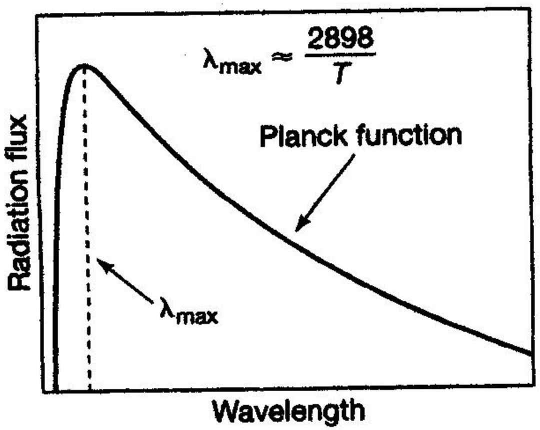
where λ_m is the WAVELENGTH in the spectrum at which the energy <u>peak</u> occurs,

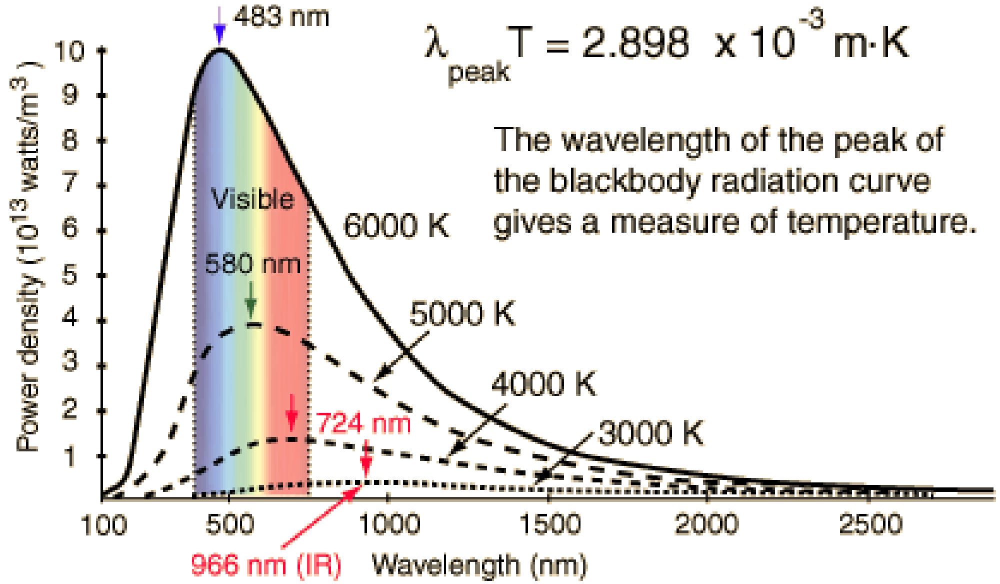
λı indicates “max”)

T is the absolute TEMPERATURE of the body, and

a is a constant (with a value of 2898)

(if λ_m is expressed in micrometers.)





- **"The wavelength of peak intensity of radiation emitted by an object is inversely proportional to its absolute temperature"**

α = 2898 (Wien's constant)

"The <u>hotter</u> the body, the <u>shorter</u> the wavelength"

"The <u>cooler</u> the body, the <u>longer</u> the wavelength"

So what Earth's radiation wavelength?

$$\lambda_{max} = a / T$$

Where:

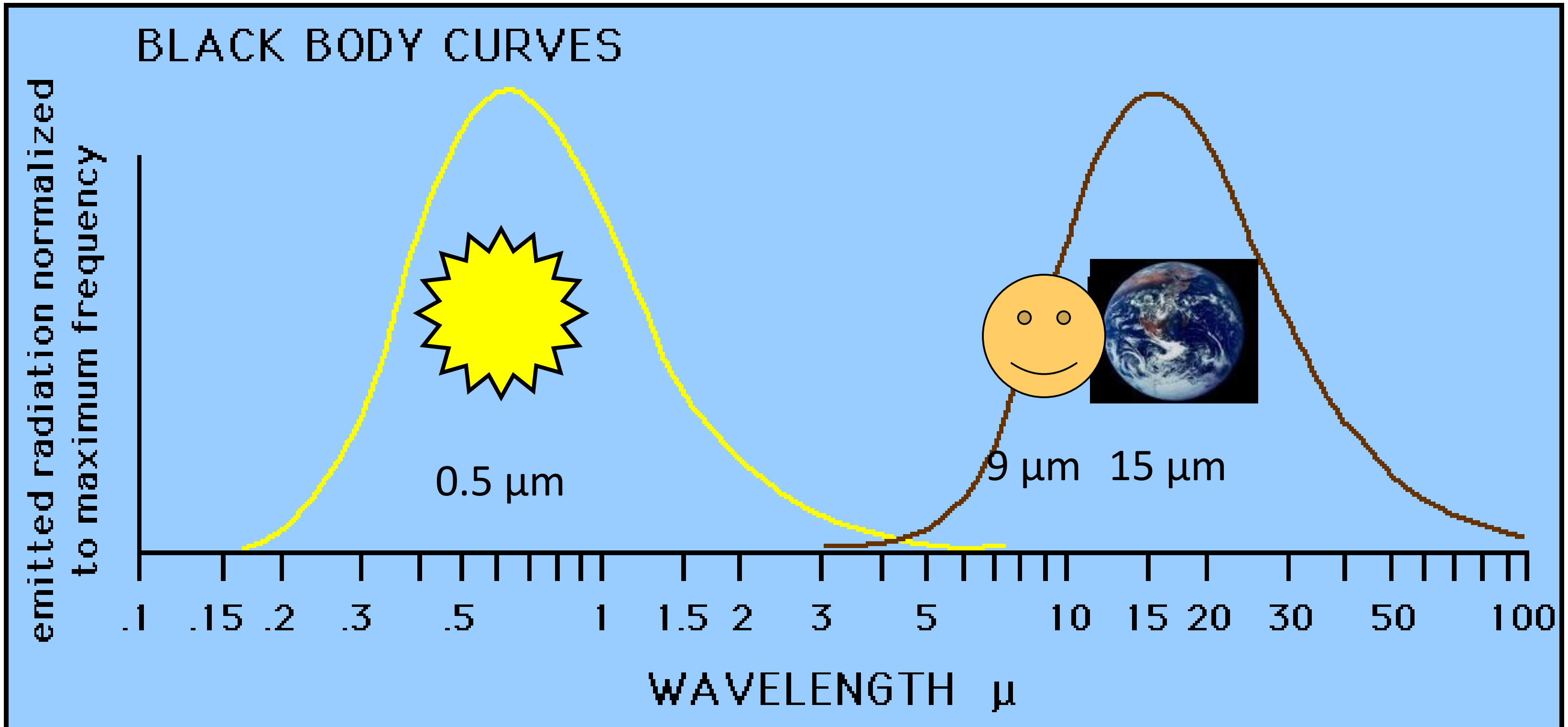
λ_{max} is wavelength of emitted radiation (in μm)

a = 2898, constant

T emitter temperature (in K)

If Earth surface temperature is 300 K
What's the wavelength?

What's the wavelength of Sun's radiation?



Range of wavelengths of solar and Earth spectra.

Emission Spectra: Sun and Earth

What's your wavelength?

$\lambda_{max} = a / T$

(a = 2898)

Your body is 37° C

or 37+273 = 310K

$\lambda_{max} = ?$

9.4 µm (far infrared)

Planck Law:

$E = h\,c / \lambda$

"SHORTER wavelengths have HIGHER intensity radiation than LONGER wavelengths"

Stefan-Boltzmann Law:

$E = \sigma T^4$

"The hotter the body, the (much) greater the amount of energy flux or radiation"

Wein's Law:

$\lambda_m = a / T$

"The <u>hotter</u> the body, the <u>shorter</u> the wavelength"

The <u>cooler</u> the body, the <u>longer</u> the wavelength"

Kirchhoff's Law of Radiation

At thermal equilibrium, the emissivity of a surface at a wavelength = its absorptivity at that wavelength.

$$\epsilon\lambda=\alpha\lambda$$

- Greenhouse gases that absorb IR (like CO_2, H_2O) also emit IR at the same wavelengths.
- Key for understanding atmospheric emission spectra.

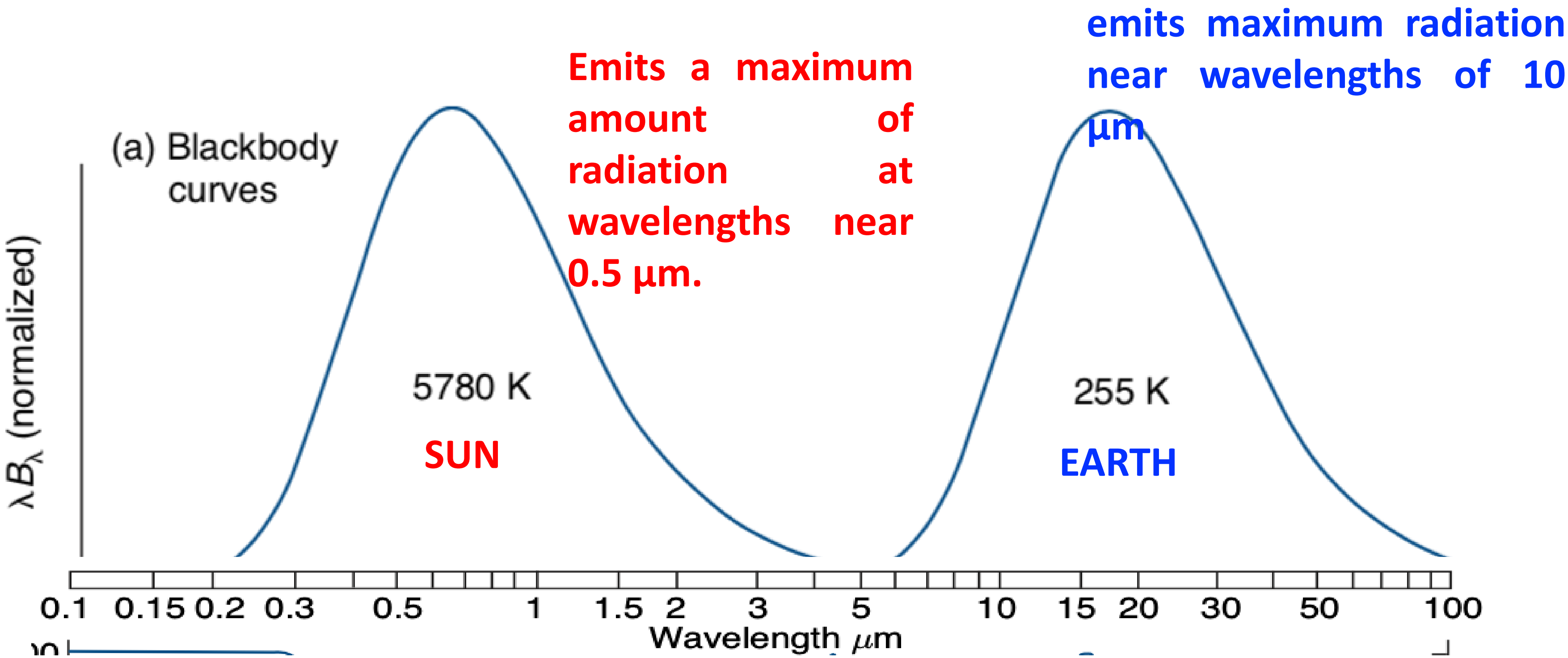
Beer–Lambert Law

Describes absorption of radiation as it passes through a medium.

$$I=I_0e^{-k_\lambda u}$$

Governs how solar and terrestrial radiation is absorbed by gases, aerosols, and clouds.

Beer–Lambert Law is the mathematical backbone of how sunlight and infrared radiation get absorbed in the atmosphere. Without it, we can't explain greenhouse warming, ozone protection, or even why the sky dims on a smoky day.

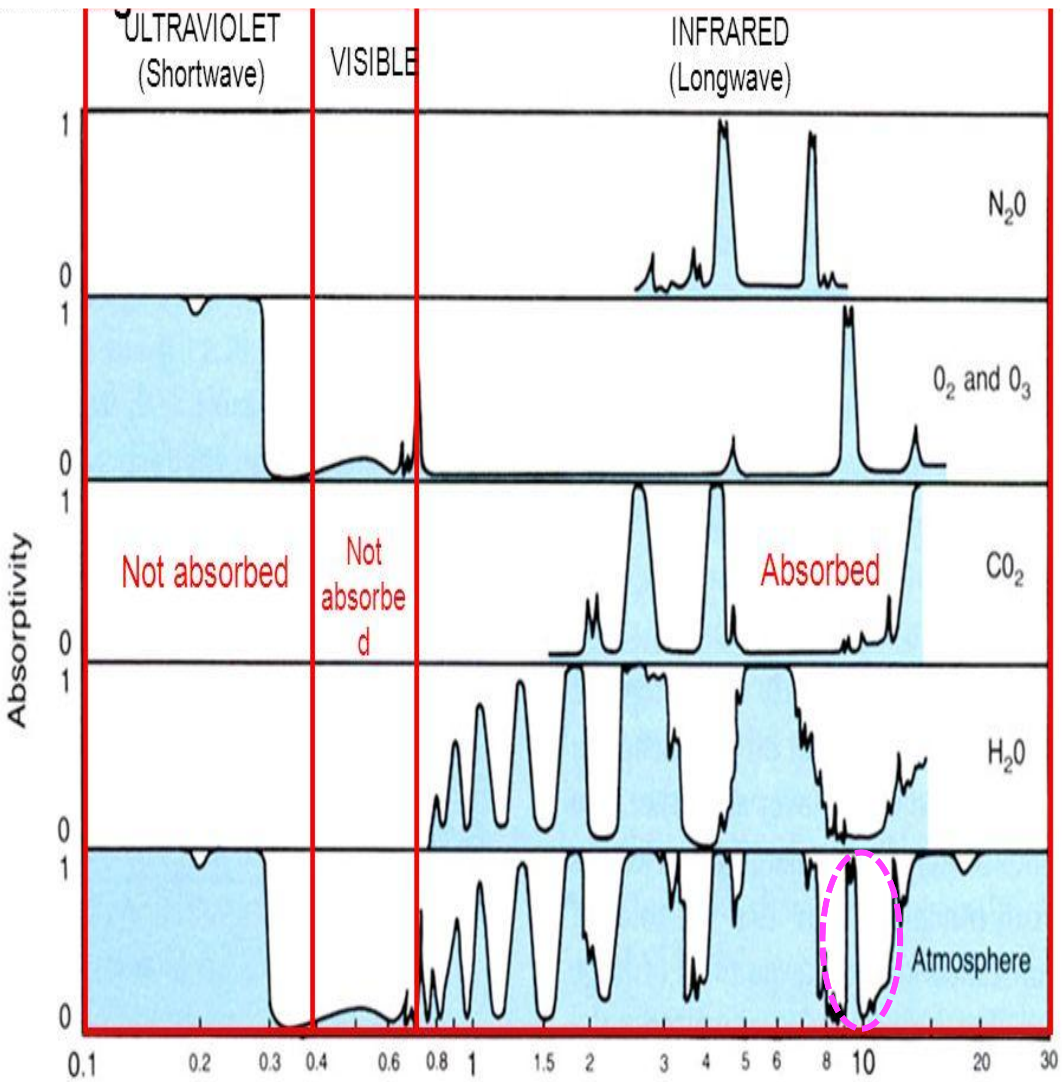


- **Wien's displacement law explains why solar radiation is concentrated in the visible (0.4–0.7 µm) and near infrared (0.7–4µm) regions of the spectrum,**

- **whereas radiation emitted by planets and their atmospheres is largely confined to the infrared (>4 µm)**

- The earth emits most of its radiation at longer wavelengths between about 5 and 25 µm, while the sun emits the majority of its radiation at wavelengths less than 2 µm. For this reason, the earth's radiation *(terrestrial radiation)* is often called **longwave radiation,** whereas the sun's energy *(solar radiation)* is referred to as **shortwave radiation.**

Absorption Spectrum: Amount of radiation absorbed at a Given wavelength



Fleagle & Businger

Absorption of solar radiation by atmosphere

- **UV radiation**
  - **Almost completely absorbed by ozone (O_3) in stratosphere**
- **Visible light**
  - **Atmosphere is largely transparent**

➢ **H_2O is the most important absorber both solar and terrestrial.**

➢ **CO_2 and Ozone come second and third**

Green house gases primarily absorb Longwave (infrared) radiation

- Both water vapor (H_2O) and carbon dioxide (CO_2) are strong absorbers of infrared radiation and poor absorbers of visible solar radiation.

- Besides being selective absorbers, water vapor and CO_2 selectively emit radiation at infrared wavelengths. This radiation travels away from these gases in all directions. A portion of this energy is radiated toward the earth's surface, thus heating the ground.

- Selective absorbers include nitrous oxide (N_2O), methane (CH_4), and ozone (O_3), which is most abundant in the stratosphere.

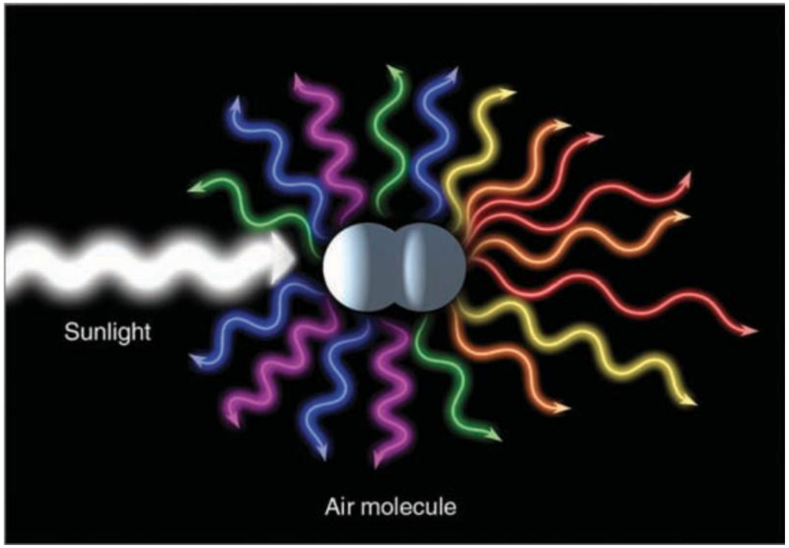
- A region between about 8 and 11 µm where neither water vapor nor CO_2 readily absorb infrared radiation. Because these wavelengths of emitted energy pass up- ward through the atmosphere and out into space, the wavelength range (between 8 and 11 µm) is known as the atmospheric window.

- Clouds can absorb the wavelengths between 8 and 11 µm, which are otherwise "passed up" by water vapor and CO_2. Thus, they have the effect of enhancing the atmospheric greenhouse effect by closing the atmospheric window.

Physics of the Scattering

when sunlight strikes very small objects, such as air molecules and dust particles, the light itself is deflected in all directions — forward, sideways, and backwards

The distribution of light in this manner is called scattering/diffusive light



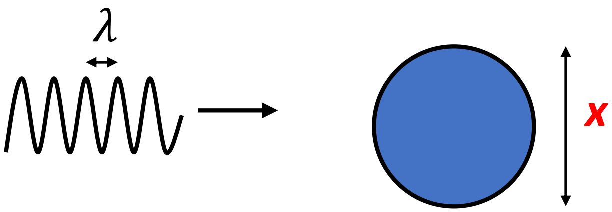
Air molecules are much smaller than the wavelengths of visible light, they are more effective scatterers of the shorter (blue) wavelengths than the longer (red) wavelengths. Hence, when we look away from the direct beam of sunlight, blue light strikes our eyes from all directions, turning the daytime sky blue

Size Parameter $x = \frac{2\pi r}{\lambda}$

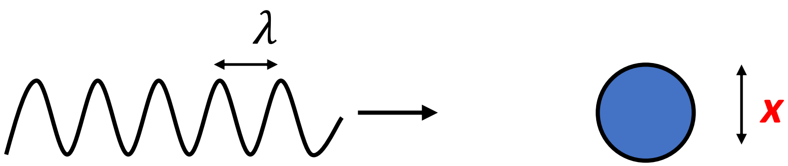
x ranges from much less than 1 for air molecules to ⍰ 1 for haze and smoke particles to >> 1 for raindrops.

- The scattering of light in the atmosphere depends on the size of the scattering particles, and on the wavelength, of the scattered light.

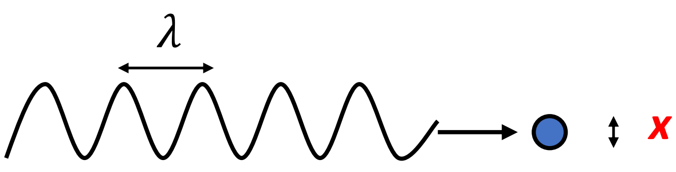
- Geometric scattering: *x* >>l
  - Rain drops (R~10-100 mm)
  - All wavelengths equally scattered
  - Optical effects: white clouds



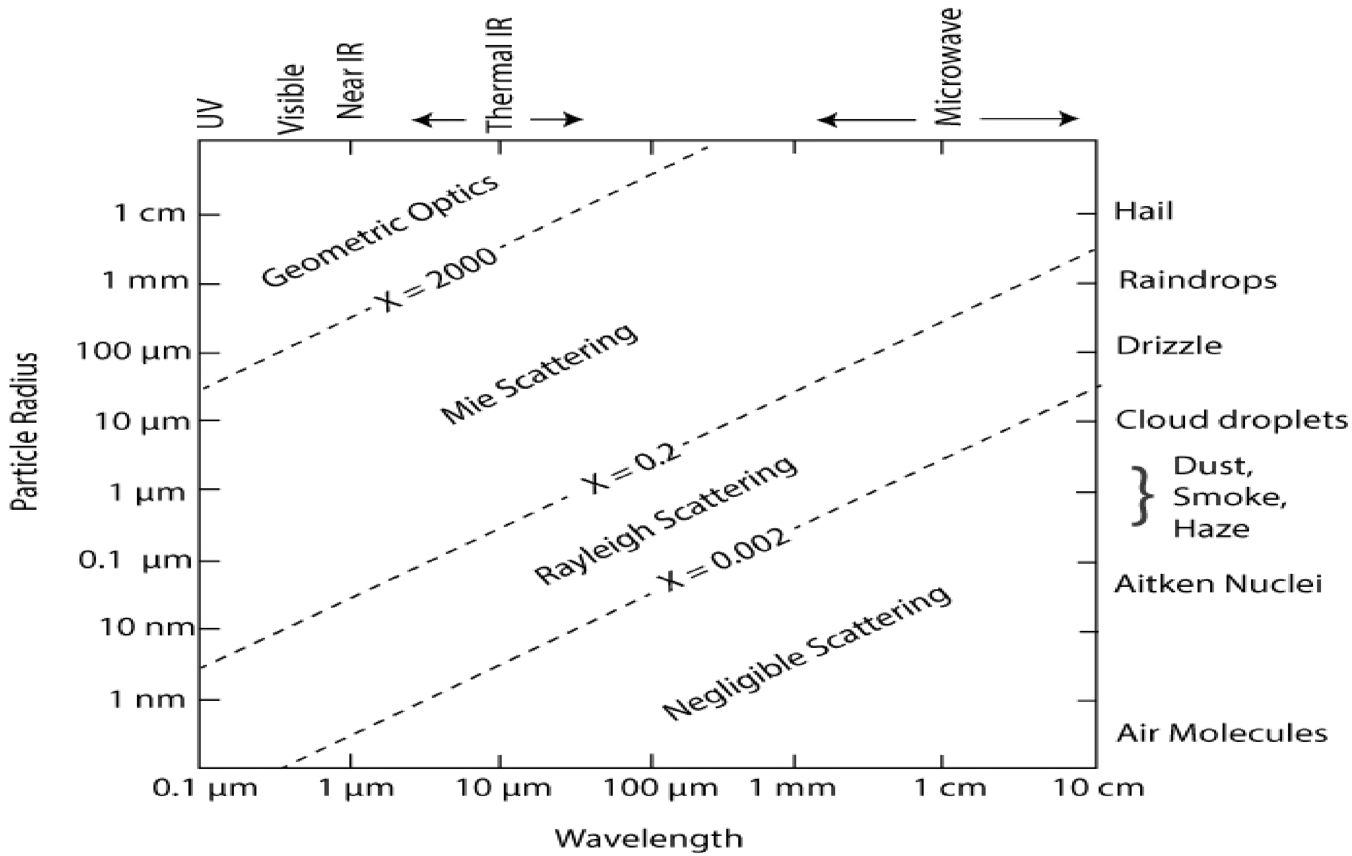
- Mie scattering: *x* ~l
  - Aerosols (R~0.01-1 mm)
  - Red scattered better than blue
  - Blue moon, blue sun



- Rayleigh scattering: *x* <<l
  - Air molecules (R~0.0001-0.001 mm)
  - Blues scattered better than red
  - red sunsets



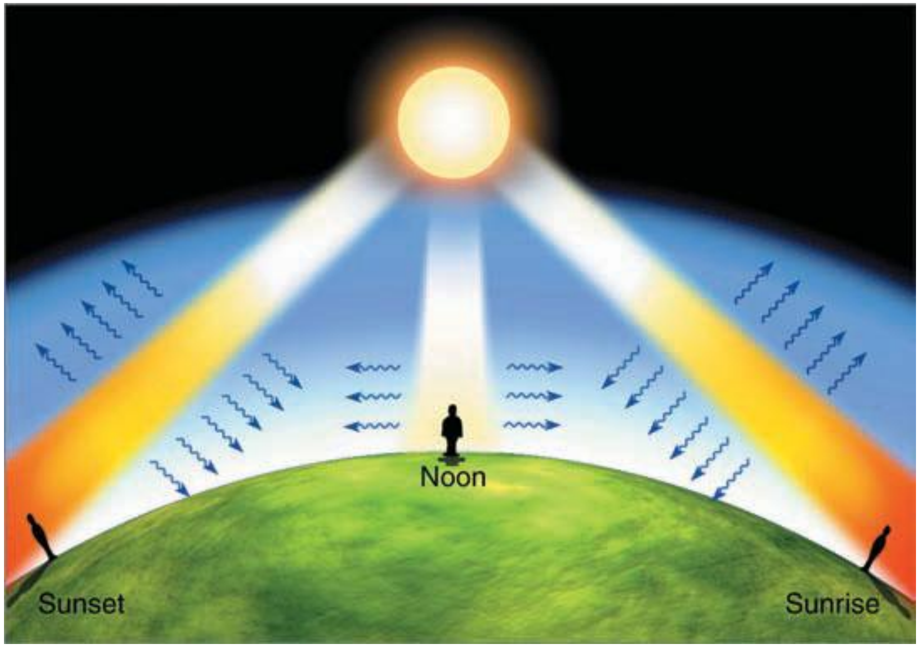
Size parameter *x* as a function of wavelength of the incident radiation and particle radius *r*



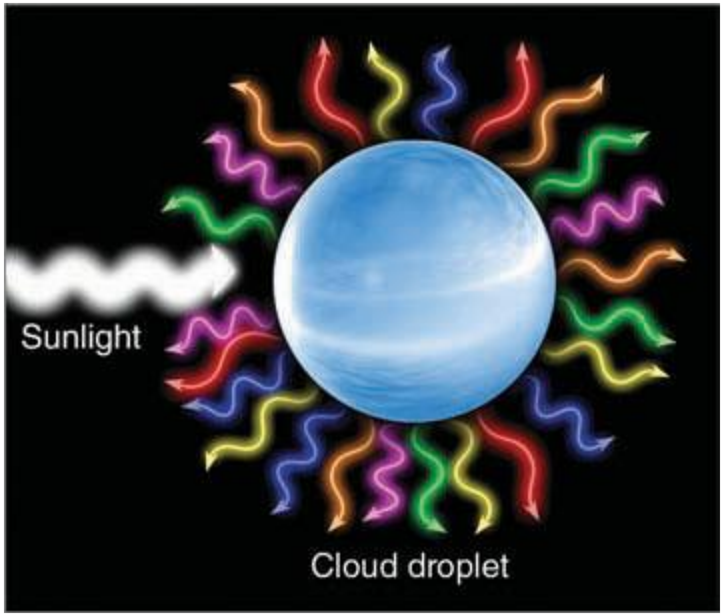
For x = 0.2 to 2000, we have **Mie scattering**. Mie scattering is when the particle size is comparable to the size of the radiation wavelength, such as many atmospheric particles and light, from UV to microwave. For x between 0.002 and 0.2, we have **Rayleigh scattering**.

Rayleigh scattering occurs for particles that are smaller than the radiation's wavelength. This is the regime for molecules for the UV into the visible and from drizzle and raindrops for radar.

Physical process:
Rayleigh scattering by air molecules and fine dust particles.



Blueness of the sky on days when the air is relatively free from aerosols



We know that the sky is blue because air molecules selectively scatter the shorter wavelengths of visible light-green, violet, and blue waves -more effectively than the longer wavelengths of red, orange, and yellow. When these shorter waves reach our eyes, the brain processes them as the color “blue.” Therefore, on a clear day when we look up, blue light strikes our eyes from all directions, making the sky appear blue.

At noon, the sun is perceived as white because all the waves of visible sunlight strike our eyes. At sunrise and sunset, the white light from the sun must pass through a thick portion of the atmosphere. Scattering of light by air molecules (and particles) removes the shorter waves (blue light) from the beam, leaving the longer waves of red, orange, and yellow to pass on through. This situation often creates the image of a reddish color sun at sunrise and sunset.

The sky is blue, but why are clouds white?

Cloud droplets are much larger than air molecules and do not selectively scatter sunlight. Instead, these larger droplets scatter all wavelengths of visible light more or less equally . Hence, clouds appear white because millions of cloud droplets scatter all wavelengths of visible light about equally in all directions.